

The Riverlabs Wari LiDAR Water Level Logger is an automatic logger system built around a Garmin LIDAR-Lite v3HP laser distance sensor. The logger is based on a low-power Atmel ATmega328P microprocessor combined with a real-time clock, 512 Kbit internal EEPROM memory, 64 Mbit flash memory and a MicroSD card reader.

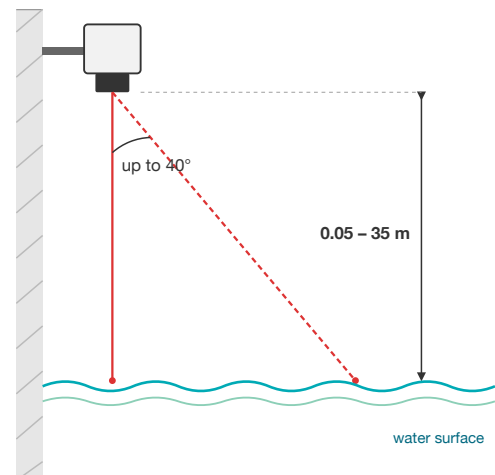
With a range of up to 35 m the LiDAR logger is suited to larger rivers, bridges and high-bank installations that are beyond the reach of ultrasonic sensors. Because the laser beam has very low divergence, the sensor can be mounted at angles of up to 40° from vertical with minimal loss of accuracy and does not require clearance from nearby banks or structures.

The firmware is fully open source and written in the Arduino programming language, so the user can adapt it to specific purposes using the freely available Arduino IDE. By default the logger measures distance to the water surface, internal temperature and battery voltage; measurements are stored in EEPROM and flushed to the SD card at regular intervals or whenever the logger is reset.



Technical Specifications

Parameter	Specification
Measurement Range	0.05 m – 35 m (dependent on target reflectivity)
Water Level Resolution	1 cm
Water Level Accuracy	~ 5 cm
Measurement Angle	Up to 40° from vertical
Temperature Resolution	0.25°C
Voltage Resolution	0.01 V
Sensor	Garmin LIDAR-Lite v3HP
Microprocessor	ATmega328P (Arduino compatible, MiniCore bootloader)
Memory	512 Kbit EEPROM + 64 Mbit internal flash + MicroSD card
Clock	Internal high-precision real-time clock
Power Supply	1 x 3.7 V Li-ion battery (18650), or 1 x 3.7 V Li-ion battery (14500) for the 5G option
Solar Charging	2 W / 6 V solar panel (cellular options)
Water Resistance	IP67



Key Features

- Long-range measurement up to 35 m – suitable for large rivers and high mounting points
- Narrow laser beam: install at angles up to 40° with no bank clearance required
- Low power consumption and long battery life
- Fully customisable open-source firmware
- Dual memory storage (EEPROM + SD)
- IP67 waterproof enclosure
- Real-time clock
- **Measured parameters:** distance to water surface, internal temperature, battery voltage

Available Options

- SD card storage only
- LoRaWAN radio module – including antenna
- 3G data connection – XBee modem
- 4G NB-IoT / LTE-M modem
- 5G modem
- Solar charging – 2 W 6 V solar panel and power regulator (included with cellular options)